

DrugTwin

Technical Documentation

Post-Inpatient TBC Compliance Companion App

Patient Companion (primary) · Photo-Verified Doses · Health Chatbot (all users) ·
Pharmacist Drug-Retention
100% Compliance + Report-Reason Escalation · SATUSEHAT-Pull · Indonesia
Version 1.3 (MVP Scope) · August 2026

Confidential — internal technical planning document.

Contents

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Consolidated technical specification for DrugTwin, built around the patient companion as the primary user. Covers product overview, account creation, architecture, tech stack, data model, API, the compliance engine (100% target, photo verification, report-reason escalation), SATUSEHAT (verified resource + practitioner linkage), AI (chatbot for all users + pharmacist drug-retention), security, and roadmap.

0. Locked Decisions

Topic	Decision
Primary user	Patient companion app (post-discharge TBC patient)
Core function	Compliance tracking: medicine + check-up reminders, progress tracker, photo-verified doses, health chatbot
Compliance target	100% for TBC — a late dose is still a miss; no 'late recovery'
Missed-dose handling	Alert patient + doctor; patient reports reason; conditional pharmacist supply notification if 'ran out'
AI scope	Chatbot for EVERY user + pharmacist drug-retention calculation
Doctor scope	Read-only dashboard + missed-dose alerts; no regimen editing (post-inpatient, fixed)
Account creation	Patient auto-created at SATUSEHAT discharge (username=NIK, default password); pharmacist/doctor self-register + admin approval
Admin role	Approves pharmacist/doctor registrations (MVP only)
Disease scope (MVP)	TBC only, extensible data models for future diseases
Tech stack	FastAPI (Python 3.14) backend + React on Bun runtime + PostgreSQL
Data source	Most data pulled from SATUSEHAT (FHIR R4); app owns photo logs + chatbot
Region / integration	Indonesia, with SATUSEHAT (national FHIR health-data exchange)
'Drug Twin' meaning	Patient's companion to stay on track with treatment and in touch with the doctor

Vision: DrugTwin is the post-discharge TBC patient's compliance companion — reminds each dose and check-up, verifies every dose with a photo, tracks progress (e.g. 1/100 days), answers health questions via chatbot, and enforces 100% compliance by alerting the doctor and escalating supply issues to the pharmacist. Patient accounts are auto-provisioned from SATUSEHAT at discharge. The 'Twin' is the patient's companion that keeps them on track and in touch with their doctor.

1. Product Overview & Scope

1.1 Problem Statement

TBC treatment in Indonesia runs for months (standard 2HRZE/4HR for drug-sensitive TB). After an inpatient stay (rawat inap), the patient continues treatment in the community. Non-adherence is the leading cause of treatment failure and drug resistance (MDR-TB), so compliance must be as close to 100% as possible. Today this is tracked manually (pill counts, paper cards), so slipping patients are caught too late.

1.2 Product Goal

A patient companion app that (1) reminds the user when to take medicine and when to attend check-ups, (2) verifies every dose with a photo as proof, (3) shows progress like 1/100 days, (4) lets the user ask health/treatment questions via a chatbot, and (5) alerts the doctor when a dose is missed. Patient accounts are auto-created from SATUSEHAT at discharge.

1.3 Users & Functions

User	Role	Account creation & functions
Pasien (Patient)	PRIMARY	Auto-created at discharge (SATUSEHAT pull; username=NIK, default password, forced change). Companion: reminders, photo dose log, progress, chatbot, report missed-dose reason.
Dokter (Doctor)	Secondary (read-only)	Self-register (Practitioner IHS + facility) → admin approves. Compliance dashboard, missed-dose alerts, chatbot. Standing SATUSEHAT link via Encounter/MedicationRequest.
Apoteker (Pharmacist)	Secondary	Self-register (Practitioner IHS + facility) → admin approves. Drug-retention, chatbot, conditional supply notification. Linked only via MedicationDispense.
Admin	Governance	Seeded (MVP). Approves pharmacist/doctor registrations (MVP only).

1.4 Account Creation & Onboarding

1.4.1 Patient — auto-created at discharge

- Triggered when the inpatient Encounter is finished ('pleased to go home').
- Backend pulls Patient + Encounter + Condition + MedicationRequest + MedicationDispense + Appointment from SATUSEHAT.
- Auto-creates User(role=patient) with username = NIK and a default password; links Patient; builds TreatmentEpisode + DoseSchedule + CheckupSchedule.
- Patient informed (SMS/letter) of NIK (username) + default password; forced password change on first login.
- Login = NIK + password.

1.4.2 Pharmacist / Doctor — self-register + admin approval

- Self-register with SATUSEHAT Practitioner IHS number + Facility(Organization); set password; status = pending.
- Admin reviews and approves → User(role) active; linked satusehat_practitioner_id + Facility.
- Login = password.

1.4.3 Admin

- Seeded account (MVP). Single MVP function: approve pharmacist/doctor registrations.
- Facility/practitioner management, patient oversight, audit, chatbot-content, config deferred to later phases.

1.5 Full Care-Plan Scope

- Medication doses (time-scheduled, photo-verified).
- Check-up / appointment days (from SATUSEHAT Appointment).
- Extensible: symptoms, weight, lab markers (sputum/X-ray) as Observations.

1.6 Non-Goals (v1)

- Not a full EMR/HIS — consumes SATUSEHAT data; does not replace the hospital system.
- No diagnosis/imaging; diagnosis & inpatient care assumed upstream in SATUSEHAT.
- Doctor does not edit the regimen in-app (post-inpatient, fixed from discharge).
- Chatbot is informational only — no autonomous diagnosis or regimen change.
- Patient self-signup / OTP onboarding — not supported; accounts are auto-provisioned from SATUSEHAT at discharge. The patient signs in with the default password and must change it on first login.

1.7 Success Metrics (proposed)

- Dose verification rate (verified ÷ scheduled) per patient.
- Median time to resolve a missed-dose reason.
- Check-up attendance rate.
- Practitioner approval latency (admin).

2. System Architecture

2.1 High-Level (Container View)

[Patient Companion App (React + Bun)] ← PRIMARY

| HTTPS / REST + JWT

[FastAPI Backend]

- Auth & RBAC (patient / pharmacist / doctor / admin)
- Account provisioning (auto patient at discharge; practitioner self-register + admin approval)
- Reminder & notification service (push/SMS/WhatsApp)
- Compliance engine (100% target, photo verification, reason/escalation)
- Chatbot service (all roles; LLM-backed, grounded)
- Photo verification & storage (object storage)
- Drug-retention service (pharmacist metric, from SATUSEHAT dispense vs photo logs)
- SATUSEHAT adapter (anti-corruption layer, FHIR)



[PostgreSQL] [SATUSEHAT API + LLM]

2.2 Layered Backend

- **API layer:** versioned routers (/api/v1/...), RBAC per route.
- **Service layer:** accounts, reminders, compliance engine, chatbot, retention, notifications, photo.
- **Repository layer:** SQLAlchemy 2.0 (async) models + sessions.
- **Integration layer:** integrations/satusehat/ maps internal models to FHIR.
- **Cross-cutting:** auth (OAuth2 + JWT), audit logging, config, PHI encryption.

2.3 Key Flows

1. Discharge intake (auto): when inpatient Encounter finishes, backend pulls Patient/Encounter/Condition/MedicationRequest/MedicationDispense/Appointment → creates patient User (NIK username, default password) + TreatmentEpisode + DoseSchedule + CheckupSchedule; notifies patient.
2. Practitioner onboarding: pharmacist/doctor self-register (Practitioner IHS + facility) → Admin approves → active.
3. Reminder: at each dose time / check-up day, notification service alerts the patient.
4. Verify: patient logs the dose with a photo → photo stored, DoseEvent verified.
5. Compliance check: engine compares verified vs scheduled; if a dose is past its deadline unverified → alert patient + doctor; patient must report a reason.
6. Supply escalation: if reason = 'ran out', conditional notification to the dispensing pharmacist (from MedicationDispense.performer).
7. Chatbot: any user asks a question → chatbot service answers from grounded knowledge (per role).
8. Pharmacist: retention service computes retained ÷ dispensed from verified DoseEvents vs SATUSEHAT MedicationDispense.
9. Sync out: adherence + observations pushed to SATUSEHAT (MedicationAdministration, Observation).

2.4 Design Principles

Patient companion is the primary surface; pharmacist/doctor/admin are derived/governance views. Internal model is disease-agnostic via `disease_type` + protocol config. SATUSEHAT coupling isolated behind an adapter. All PHI access audited. Alert routing follows the real SATUSEHAT practitioner linkage (doctor = standing; pharmacist = dispense event).

3. Technology Stack

3.1 Backend

- Python 3.14, FastAPI, Pydantic v2, SQLAlchemy 2.0 (async), AsyncPG, Alembic, Uvicorn.
- Auth: OAuth2 password + JWT (PyJWT), bcrypt. RBAC: patient/pharmacist/doctor/admin. Patient login = NIK + password.
- Notifications: patient = in-app only; doctor & pharmacist = in-app + WhatsApp Business API. SMS still used for discharge credentials + default-password delivery.
- Photo storage: object storage (S3-compatible); timestamped, access-controlled.
- Chatbot: LLM via provider-agnostic wrapper, grounded retrieval over treatment/FAQ/retention content; no PHI leakage.
- SATUSEHAT client: httpx (async), OAuth2 client-credentials, FHIR R4.
- Testing: pytest + httpx TestClient; TBC fixtures + negative (non-TB) intake-guard cases.

3.2 Frontend (Patient Companion)

- React + TypeScript, Bun runtime/package manager, Vite.
- Camera capture for dose-photo verification; push notifications; offline-friendly logging queue.
- TanStack Query + fetch; Zustand state; Tailwind; Recharts for progress; React Hook Form + Zod.
- i18n: id (Bahasa Indonesia) primary, en fallback. Mobile-responsive (companion app first).

3.3 Infra / Ops

- PostgreSQL 16. Docker Compose (api + web + db + optional minio).
- 12-factor env; secrets in secret manager; never committed.
- CI: ruff/black + eslint/prettier + type-check + tests.

4. Data Model

Extensible: disease_type + protocol config drive TBC today; new diseases = config rows, not schema changes. Reference data is pulled from SATUSEHAT; the app owns

DoseEvent photos and chatbot conversations. The patient User is auto-created at discharge with username = NIK.

4.1 Entity Overview (ERD)

Entity	Key Fields	Purpose / SATUSEHAT source
User	id, username(=NIK for patient), email, hashed_password, role, status, force_pw_change	Auth & role (patient/pharmacist/doctor/admin)
Patient	id, user_id?, satusehat_patient_id, nik(masked), name, dob, phone	Person (SATUSEHAT Patient)
Practitioner	id, user_id, satusehat_practitioner_id, facility_id, status(pending/active)	Doctor/pharmacist (SATUSEHAT Practitioner)
Facility	id, satusehat_organization_id, name, type	Clinic/pharmacy (SATUSEHAT Organization)
Encounter	id, patient_id, satusehat_encounter_id, period, status(finished)	Inpatient → discharge (SATUSEHAT Encounter)
TreatmentEpisode	id, disease_type, total_days, target_compliance=100, status	TBC episode (extensibility hinge)
MedicationRegimen	id, episode_id, protocol_key, drugs, doses_per_day	From SATUSEHAT MedicationRequest
MedicationDispense	id, request_id, performer_practitioner_id, quantity	SATUSEHAT MedicationDispense (pharmacist link)
DoseSchedule	id, regimen_id, scheduled_at, window_min, grace_min	Derived from dosageInstruction
DoseEvent	id, schedule_id, photo_url, verified, status, reason, reported_at	Photo-verified intake (app-owned)
CheckupSchedule	id, episode_id, scheduled_date, type, status	SATUSEHAT Appointment
ChatbotMessage	id, user_id, role, content, citations, ts	Conversation (app-owned)
Observation	id, episode_id, type, value, effective_at	Weight/sputum/symptom (SATUSEHAT

		Observation)
Alert	id, episode_id, type, reason, route(doctor pharmacist), status	Missed-dose / supply escalation
AuditLog	id, actor_id, action, resource, phi_accessed, ts, ip	PHI audit trail

4.2 Progress, Compliance & Retention Math

- **Progress:** $\text{verified_days} / \text{total_days}$ (streak). Example: 1/100 for consuming medicine 100 days straight.
- **Compliance %:** $\text{verified doses} \div \text{scheduled doses}$ (TBC target = 100%; a late dose is still a miss).
- **Drug retention (pharmacist):** $\text{retained_doses} \div \text{dispensed_quantity}$, computed from verified DoseEvents vs SATUSEHAT MedicationDispense.

5. API Design

Versioned under /api/v1. Auth: POST /auth/token → JWT; Bearer on all others. Patient login = NIK + password; practitioner/admin login = password. RBAC per route. Errors RFC7807.

Pagination limit/cursor.

Group	Endpoint	Purpose
Auth	POST /auth/token	Login (patient: NIK+password; practitioner/admin: password)
Auth	POST /auth/change-password	Forced change on first login / reset
Auth	GET /me	Current user + role
Practitioner	POST /auth/register/practitioner	Self-register (Practitioner IHS + facility), pending
Admin	POST /admin/approve/{user_id}	Approve practitioner registration (MVP admin function)
Patient	GET /my/today	Today's doses + check-ups
Patient	POST /doses/{id}/verify	Log dose + photo (verified)
Patient	GET /my/progress	$\text{verified_days} / \text{total_days}$ + compliance %
Patient	POST /chat	Ask health question (chatbot)
Patient	GET /my/checkups	Upcoming check-ups
Patient	POST /alerts/{id}/reason	Report missed-dose reason
Doctor	GET /episodes/{id}	Compliance dashboard

		(read-only)
Doctor	GET /alerts?route=doctor	Missed-dose alerts from patients
Pharmacist	GET /patients/{id}/retention	Drug-retention metric
Pharmacist	POST /chat	Ask retention/patient-history question (chatbot)
Pharmacist	GET /alerts?route=pharmacist	Conditional 'ran out' supply notifications
SATUSEHAT	POST /integrations/satusehat/sync/patient/{id}	Pull intake at discharge
SATUSEHAT	POST /integrations/satusehat/push/episode/{id}	Push adherence/Observation

6. Compliance Engine (TBC)

6.1 100% Compliance Target

TBC requires zero missed doses. A late dose is still a miss — there is no 'take it late to recover' path; the streak/compliance stays broken. The engine treats any unverified scheduled dose past its deadline as a failure.

6.2 Photo Verification

Every dose must be confirmed with a photo as proof. DoseEvent is created with photo_url + verified=true only on successful on-time submission; timestamp recorded for evidence.

6.3 Missed-Dose → Alert → Report Reason (no late recovery)

10. A scheduled dose is not verified by its deadline → engine raises a missed-dose Alert to the patient AND the doctor (doctor has the standing SATUSEHAT link).
11. The patient must take a mandatory 'next step': report a reason (ran out / side effect / forgot / other). There is no 'take it late' recovery — compliance is already broken.
12. If reason = 'ran out' (supply issue) → conditional supply notification to the dispensing pharmacist, identified from SATUSEHAT MedicationDispense.performer.actor.

6.4 Progress & Retention

- Progress shown as verified_days / total_days (streak), e.g. 1/100.
- A missed/unverified dose breaks the streak; compliance must be restored toward 100%.
- Pharmacist drug-retention = retained ÷ dispensed, computed continuously from verified DoseEvents vs SATUSEHAT MedicationDispense.

6.5 Human-in-the-Loop

The engine alerts and escalates; the doctor monitors and reviews (no regimen editing in-app). It never auto-discontinues treatment. Protocol definitions live in config so new diseases reuse the engine.

7. SATUSEHAT Integration

7.1 Why

SATUSEHAT is Indonesia's national health information exchange (FHIR R4). It supplies the reference data the app needs (including the Patient record used to auto-provision the patient account at discharge); the app adds photo-verified intake and chatbot conversations, and pushes adherence back.

7.2 Resource Coverage (verified)

App data need	SATUSEHAT resource	Notes
Patient demographics (NIK, name, DOB)	Patient	search by NIK / name / DOB; NIK = patient username
Inpatient encounter / discharge	Encounter	rawat inap class, period; triggers account creation at finished
TBC diagnosis	Condition (ICD-10)	linked to Encounter
Medication regimen + schedule	MedicationRequest	dosageInstruction → derive DoseSchedule
Pharmacy dispensing	MedicationDispense	powers pharmacist drug-retention
Check-up / appointment	Appointment	entered by hospital → SATUSEHAT; pulled by app
Weight / sputum / labs	Observation (LOINC)	progress markers
Clinician / facility	Practitioner / Organization	practitioner self-register + admin approval

7.3 Practitioner Linkage & Alert Routing

- **Patient ↔ Doctor (standing):** Encounter.participant.individual → Practitioner (attending) and MedicationRequest.requester → Practitioner (prescriber). Doctor legitimately receives the missed-dose alert.
- **Patient ↔ Pharmacist (event-based only):** MedicationDispense.performer.actor → Practitioner (dispenser); authorizingPrescription → MedicationRequest; subject → Patient. No standing care-team relationship.
- **No CareTeam resource:** doctor + patient + pharmacist are traceable via Encounter/MedicationRequest/MedicationDispense, not a shared care team.

- **Routing rule:** missed-dose alert → doctor (standing link); pharmacist gets only a conditional supply notification when reason = 'ran out', routed via MedicationDispense.performer.

7.4 Strategy

- Pull intake at discharge
(Patient/Encounter/Condition/MedicationRequest/MedicationDispense/Appointment); push adherence + observations asynchronously (idempotent via external IDs).
- Anti-corruption layer isolates FHIR quirks; internal model stays disease-agnostic.
- Sandbox first; production credentials via secret store. Consent & data-minimization; PHI audit on sync.

8. AI Decision Support

8.1 Chatbot — for EVERY user

A conversational assistant available to all roles, grounded in their relevant data: patient asks treatment/health questions; pharmacist asks about retention/patient history; doctor asks about compliance. Informational only; cites sources; escalates clinical questions to a human.

8.2 Pharmacist Drug-Retention

From photo-verified DoseEvents vs SATUSEHAT MedicationDispense, compute how much of the dispensed regimen the patient has actually retained/taken. Surfaces as a metric for refill planning and adherence flagging. This is a calculation/metric, not an autonomous decision.

8.3 Guardrails

- Chatbot informational only; explicit human-in-the-loop for any clinical action.
- No PHI leaves the boundary unchecked; answers cite source data.
- LLM behind a swappable interface; log prompts/decisions for audit. Scope limited to adherence & monitoring.

9. Security & Privacy

- **Regulatory:** Indonesia UU PDP — lawful basis, consent, minimization, breach notification.
- **PHI:** encryption at rest (TDE / column crypto for NIK + photos), TLS in transit; NIK masked/tokenized at rest (used as login username).
- **Photo proof:** stored access-controlled, timestamped, retained per policy; patient-consented. Store-only — no AI check that the photo shows the medicine.
- **AuthN/Z:** OAuth2 + short-lived JWT + refresh; RBAC — patient sees own; pharmacist/doctor see assigned cohort; admin sees pending approvals.

- **Default password:** forced change on first login; delivered over SMS/letter, not stored in plaintext.
- **Audit:** every PHI read/write logged (actor, resource, ts, ip), incl. SATUSEHAT MPI resolve at discharge.
- **Secrets:** SATUSEHAT + LLM credentials in secret manager; never in VCS.
- **Validation:** Pydantic on inputs; rate limiting; secure headers; per-role authz tests.

10. Delivery Roadmap

Phase	Focus	Key Output
0 — Scaffold	Rebuild base	Repo, Docker Compose, CI, Postgres, auth+RBAC, models, Alembic
1 — Companion + accounts	Remind + verify + provision	SATUSEHAT discharge pull → auto patient account (NIK+default pw); practitioner self-register + admin approve; reminders, photo-verified dose log, progress X/total
2 — Strict compliance	100% + escalation	Missed-dose alert→doctor, report reason, conditional pharmacist supply notification, drug-retention metric
3 — Chatbot	All-role Q&A	Grounded chatbot for patient/pharmacist/doctor
4 — SATUSEHAT	Interop harden	Push MedicationAdministration/Observation/Appointment; sandbox→prod
5 — Hardening	Pilot	Security/PDP review, audit coverage, UAT, doc finalization

Definition of done per phase tracked as issues. CI gates: lint + type-check + tests.

10.1 Target Repository Structure

Drug Twin/

- backend/
app/{main,api/v1,core,models,schemas,services,integrations/satusehat,db}, tests/, pyproject.toml, Dockerfile
- frontend/ src/{api,features,components,pages}, package.json (bun), vite.config.ts, Dockerfile
- DrugTwin-Technical-Docs.docx
- docker-compose.yml

- README.md

Appendix A. Assumptions & Open Questions

A.1 Assumptions

- TBC regimen follows standard Indonesia guidelines; total_days configurable (user example: 100).
- 'Drug Twin' = patient's companion to stay on track with treatment and in touch with the doctor (confirmed).
- Check-up dates entered by hospital, fed into SATUSEHAT Appointment, then pulled by DrugTwin (confirmed).
- Photo proof stored as timestamped evidence; optional on-device/AI check of medicine presence (future).
- Web/responsive companion app for v1; native mobile later. Admin account seeded for MVP.

A.2 Resolved Decisions

13. Patient companion is primary user.
14. Compliance = full care plan; 100% target for TBC (late = miss, no recovery).
15. AI = chatbot for all users + pharmacist drug-retention.
16. Missed dose → alert doctor; patient reports reason; 'ran out' → conditional pharmacist supply notification.
17. Doctor read-only (no regimen editing); pharmacist linked only via MedicationDispense.
18. Most data pulled from SATUSEHAT (verified resource + practitioner linkage).
19. Patient account auto-created at SATUSEHAT discharge (username=NIK, default password, forced change).
20. Pharmacist/doctor self-register + admin approval; Admin MVP function = practitioner approval only.

A.3 Resolved (post-MVP clarifications)

- 'Drug Twin' = patient's companion to stay on track with treatment and in touch with the doctor.
- Check-up dates entered by hospital, fed into SATUSEHAT Appointment, then pulled by DrugTwin (not in-app by patient).
- Photo verification = store timestamped proof only; no AI check that the photo shows the medicine.
- Notifications: patient = in-app only; doctor & pharmacist = in-app + WhatsApp (SMS used for discharge credentials).

All previously open items are now resolved; the MVP spec is fully locked.